Ontypology x Mind AI

AI Gene R&D Brain

Gene R&D Brain

The complete solution for accurate output usage

Gene therapy and gene editing R&D x CRISPR delivery case



Type = Type, Space Type x Time Type, Mind AI platform

MIND AI × ONTYPOLOGY

Accurate, complete, provable

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Before we begin: three conventions

This is a "how to use it" book. The AI gene R&D brain writes every output of gene therapy and gene editing R&D - target card, guide design, vector plan, assessment report, R&D verdict, evolution record - as a "Type" with fields, criteria, and identity, so that every step of R&D is reasonable, acceptable, and recheckable. Before picking up the pen, establish three conventions.

Convention one Type = Type, not Form

The English for the character for Type throughout this book is always Type - SpecType, GuideType, Type-Driven R&D. Not Form.

Convention two the theory is used, not taught

The Space Type and the Time Type are the foundation of this book. The book uses only the tools of the theory - the three brackets [ ] { } ( ), the intersection criterion ∩(T), the identity id(C), the three levels of remove, the four framework steps, the three-layer judgement, and the fourteen Time moves - without restating the principles and arguments. To read the theory itself, return to the originals "The Space Type of Ontypology" and "Ontypology: The Time Type".

Convention three the platform is cited, not restated

Mind AI is the underlying platform of this book: the LLM handles inductive statistics, the Mind Model handles deductive reasoning, and the two converge in the platform. The book cites this division of labour without restating the platform; to understand the platform itself, return to "Introduction to Mind AI".

Demonstration declaration: the programme code, sequence parameters, efficiency, and off-target values of the running case GX-2610 (HBB gene correction for sickle-cell anaemia) are all demonstration data,
Used to demonstrate the output format and the operation of the criteria; it does not constitute medical advice, nor is it a basis for real clinical trial design. Real gene-editing R&D must follow regulations and ethics review (see 6.5).

Table 0-1 Task map: how this book takes you through

Chapter	Topic	Corresponding tool	What you will get
Before we begin	Three conventions and the demonstration declaration	Type = Type, used not taught	How to read the book
Guided reading	Two languages, three ways of reading	The coordinates of the two Types	The panorama in one figure
Chapter 1 (understanding the system)	Understanding the R&D brain, the four scenes of inaccurate output	Type + Number, the move tray	Why a Type is needed
Chapter 2 (overview of the six Types)	Target, guide, vector, assess, verdict, evolution: six-Type field cards	The three brackets, the three levels of remove	The template of the output
Chapter 3 (the four gates of accuracy)	Criteria, skeleton, acceptance, trace	∩(T), Type evidence chain	The definition of accuracy
Chapter 4 (the six operating steps)	The operation from R&D requirement to output	Four framework steps, three-layer judgement	Operation card
Chapter 5 (the GX-2610 case)	Full output example of the seven-stage relay	Full text of prompts S1–S7	A template that can be copied
Chapter 6 (misuse and maturity)	Misuse comparison, L1–L4, compliance red lines	The ten-question checklist, adoption blueprint	Adoption route
Review	Three takeaways	Toolbox	Carry-along card

Chapter	Topic	Corresponding tool	What you will get
Appendix A (quick reference)	Terminology, moves, notation	Quick-reference table	Look it up as you go

Guided reading: two languages and three ways of reading

Gene R&D has one language: target, sgRNA, vector, off-target, IND. Ontypology has another: Type, three brackets, criteria, moves. The two languages speak of the same thing - this book is their translation manual.

Table D-1 Two languages compared

The language of Ontypology	The language of the R&D site	State it clearly in one sentence
Type	Specification, design document, report	Output with fields and criteria
[ ] property, abstraction	Target features, design conditions	Dimensions and scales that can be questioned
{ } BOM, composition	System composition, formulation, module list	Open it up and see what it is made of
( ) Container, identity	Programme, batch, cell line	A frame whose content changes but whose identity does not
∩(T) intersection criterion	Acceptance thresholds, release standards	Only when all pass does it count
three levels of remove	Part impact grading	Whether identity survives after removal
Move (function)	R&D operating method	Time Type = verb = function
Type evidence chain	R&D records and audit trail	Every step recheckable

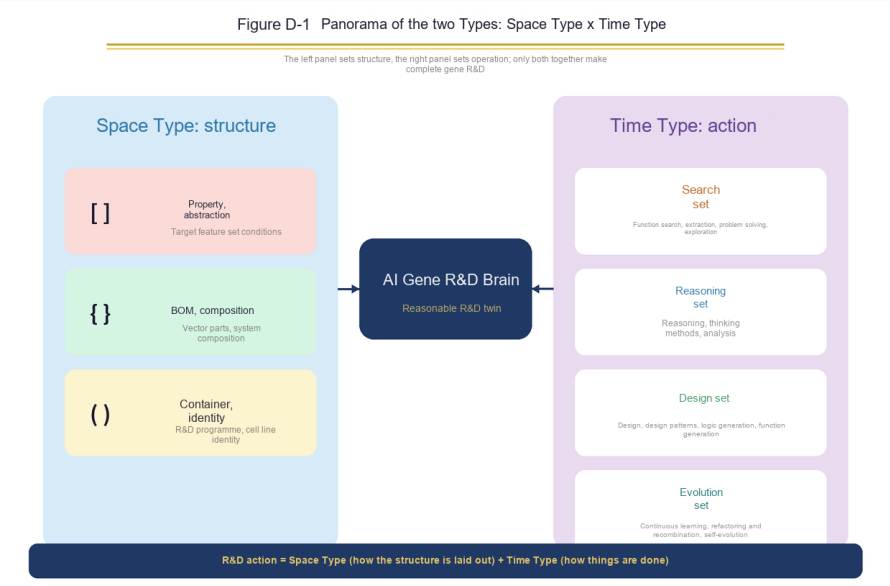


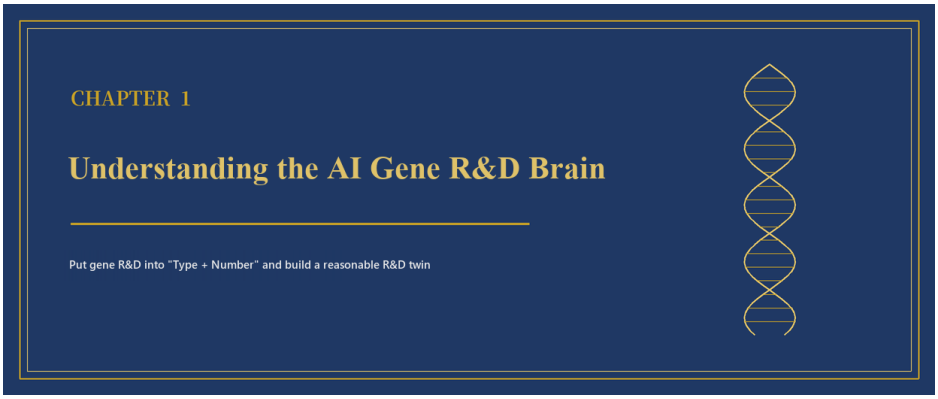
Figure D-1 Panorama of the two Types: the Space Type sets structure, the Time Type sets operation

Table D-2 Three ways of reading

Who you are	Suggested path	Estimated time
Executive, decision maker	Before we begin, then 1.4, then 3.5, then 5.4, then 6.3	30 minutes
R&D engineer	Guided reading, then 2, then 4, then all of chapter 5	2 hours
Adoption driver	1 → 2 → 4 → 6	90 minutes

Path reading: the numbers point to chapters - the management line looks at decisions, the engineer line looks at output, the adoption line looks at process.

Chapter One Understanding the AI Gene R&D Brain



The AI gene R&D brain is not a chatbot but a "reasonable R&D twin" - it packs the knowledge, criteria, and decisions of gene therapy and gene editing R&D into a frame built of Type and Number.

The difficulty of gene R&D is not that data cannot be produced, but that there is always a gap between data and decision: literature sits in folders, parameters sit in engineers' heads, decisions sit in meeting minutes. By the time you need to recheck, hand over, or explain to a regulator, nobody can piece the reasoning back together. The AI gene R&D brain's solution is to write R&D itself as a Type - every output has fields, every gate has criteria, every decision has a documented basis.

1.1 The Reasonable R&D Twin: How Type + Number Holds Gene R&D

Brain thinking = Type + Number. Number answers "how much, how fast, how accurate" - editing efficiency 68%, off-target rate 0.3%, cell survival 86%; Type answers "what it is, what it is made of, how it changes" - does this pegRNA belong to the Guide Type, what does this delivery system consist of, and is this programme still the same one after changing a batch of cells. R&D with numbers alone yields a pile of numbers but no reasoning; only Type and Number together yield an R&D twin that can be questioned, rechecked, and handed on.

The core carrier of this twin is the "Type" (Type, not Form). Every output of R&D - target card, guide design, vector plan, assessment report, verdict document, evolution ledger - is a Type: with fields (properties), criteria (∩(T)), identity (id(C)), and composition (BOM). Only when the output is written as a Type does AI have a skeleton for reasoning and people have a target for acceptance.

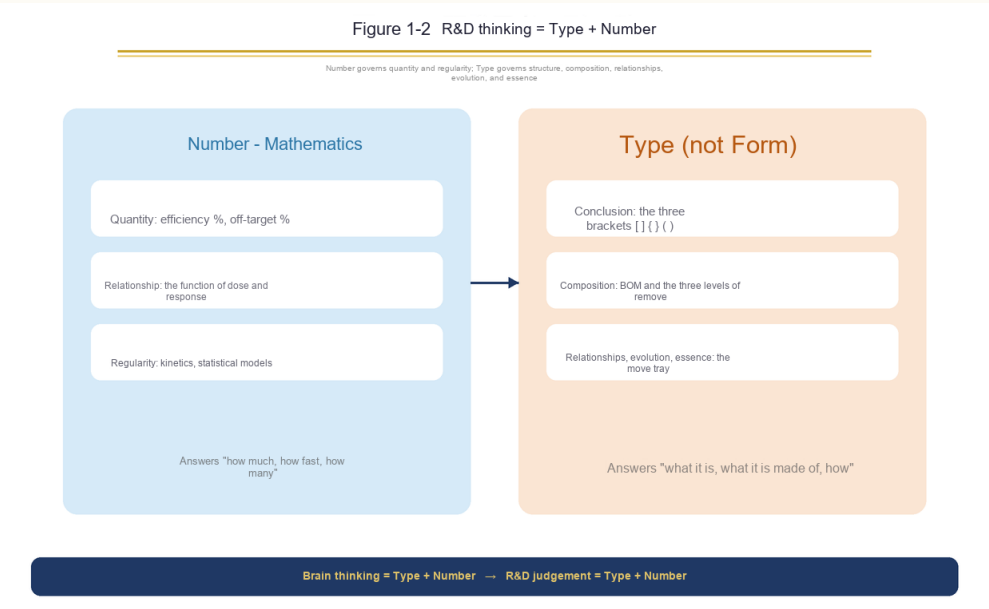


Figure 1-2 R&D thinking = Type + Number

R&D twin = Type (structure, composition, evolution) + Number (quantity, relationship, regularity)

1.2 Where the Two Types Land: the Three Space Brackets and the Fourteen Time Moves

The Space Type governs "how the structure is arranged": the three brackets each write one layer - [ ] writes class rules, { } writes composition structure, ( ) writes the identity container. The Time Type governs "how things are done": the Time Type is verb, function, move; every act of R&D is selecting, assembling, and swapping moves on the fourteen-move tray.

Table 1-1 The three Space brackets landing in gene R&D

Bracket	Ontypology meaning	Where it lands in R&D	Instance
[ ]	Property, abstract set	Target features, design conditions, acceptance thresholds	F(Guide Type) = { complementary target sequence, qualified PAM, editing window covering the variant }
{ }	BOM, composition set	System composition, formulation, module list	BOM(LNP vector) = { lipid, mRNA, PEG lipid, targeting ligand }
( )	Container, spatial container	The identity of programme, batch, and cell line	C = (id(C), S(C)): when GX-2610 changes cell batch, id(C) does not change

The fourteen moves of the Time Type fall into four sets: the search set finds it, the reasoning set thinks it clearly, the design set makes it, the evolution set grows it. The R&D situation decides which move to pick - designing pegRNA picks "design + generation", when efficiency will not come up pick "refactoring", when the assumption is unstable pick "first-principles thinking". The fourteen moves are not a checklist but a move tray that moves with the situation.

Table 1-2 The fourteen Time moves x where they land in gene R&D

Set	Move	Where it lands in R&D
Search set	Type-time function search	Find reusable designs in the Type library: past pegRNA parameters, vector formulations
Search set	Design search and extraction	Extract editing strategies and sequence design space from literature and patents
Search set	Problem solving	Troubleshooting path for efficiency bottlenecks and off-target anomalies

Set	Move	Where it lands in R&D
Search set	R&D and exploration	Feasibility exploration of new editors and new delivery routes
Reasoning set	Reasoning	Causal, reverse-causal, simulation, statistical, reverse: infer mechanism from data, infer variant from phenotype
Reasoning set	Thinking methods	Forward, reverse, horizontal, vertical, first principles: strategy divergence and convergence
Reasoning set	Analysis	Sequence analysis, off-target profile analysis, SAR-style structure-activity analysis
Design set	Design (creation and innovation)	Design of pegRNA, vector, and validation plan
Design set	Design patterns	Reusable R&D patterns: screening funnel, control design, dose escalation
Design set	Logic generation and application	Generate the criteria as executable acceptance rules
Design set	Function generation and application	Seal the process into a repeatably callable standard operation
Evolution set	Continuous learning	Each round's experimental data feed back into the Type library
Evolution set	Refactoring and recombination	PBS/RTT combination rearrangement, strategy refactoring and re-measurement
Evolution set	Self-evolution	Criteria thresholds update with the evidence, and the Type library grows by itself

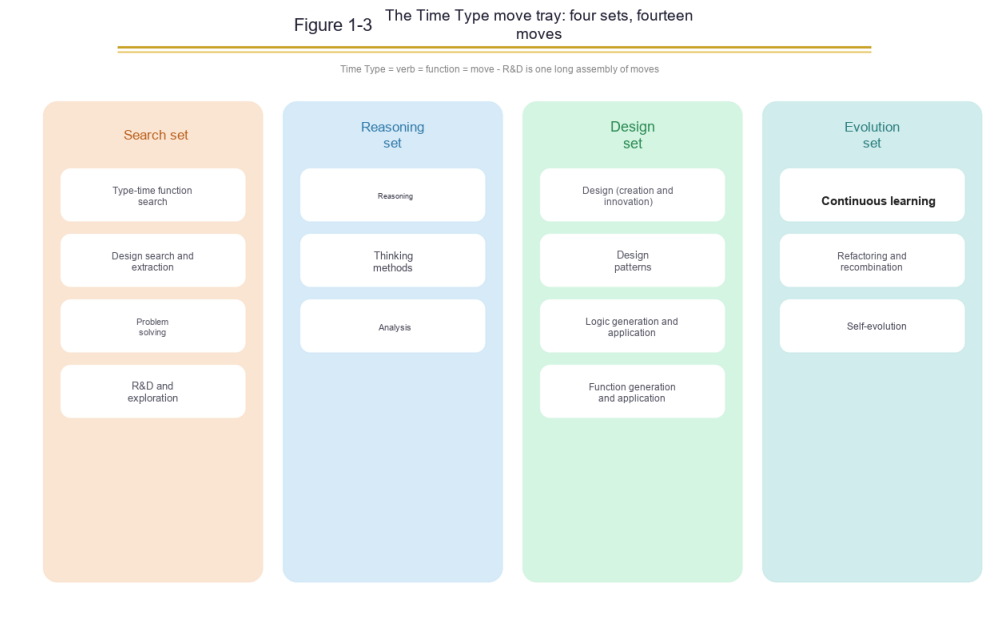


Figure 1-3 The Time Type move tray: four sets, fourteen moves

1.3 The Mind AI Division of Labour: LLM Induces, Mind Model Deduces

The underlying platform is Mind AI. The division of labour in one sentence: the LLM handles language ability, the Mind Model handles intelligence. In gene R&D - the LLM reads literature, writes reports, recalls answers (inductive statistics); the Mind Model reasons, accepts, remembers, evolves (deductive reasoning). Accurate output does not come from a bigger model but from each doing its own job with the criteria refereeing in between.

Table 1-3 The Mind AI division of labour in gene R&D

Role	Good at	What is done at the R&D workflow
LLM (inductive statistics)	Language understanding, language generation, knowledge recall	Read literature and extract, write the first draft of each Type's output, organise experimental records

Role	Good at	What is done at the R&D workflow
Mind Model (deductive reasoning)	Unified knowledge, deep reasoning, long-term memory, autonomous learning	Bring the first draft into the Type skeleton, run the ∩(T) criteria, and maintain the Type evidence chain and evolution ledger
R&D staff	Situation judgement, value trade-off, final sign-off	Give the situation, review the output, give the verdict - people handle direction, the Type handles clarity

1.4 Where Accuracy Comes From: Preview of the Four Gates

"Accurate output" is not a slogan; it is gated out by four gates: first set the criteria (the ∩(T) funnel), then deduce the skeleton so the output grows on the skeleton, then reconcile with the experimental data three-piece set, and finally string every step into a Type evidence chain. Only when all four pass is it accurate; if any one fails, return - and the return also leaves a trace.

Table 1-4 Preview of the four gates

Gate	One sentence	What to output	Detail
Gate one the criterion funnel	Set the criteria first, then let the candidates pass	∩(T) acceptance sheet	3.1
Gate two deductive skeleton	The output grows on the skeleton, not on feeling	Deductive skeleton	3.2
Gate three validation reconciliation	When the experimental data come back, reconcile item by item	The validation three-piece set	3.3
Gate four trace into a chain	Every step recheckable and replayable	Type evidence chain	3.4

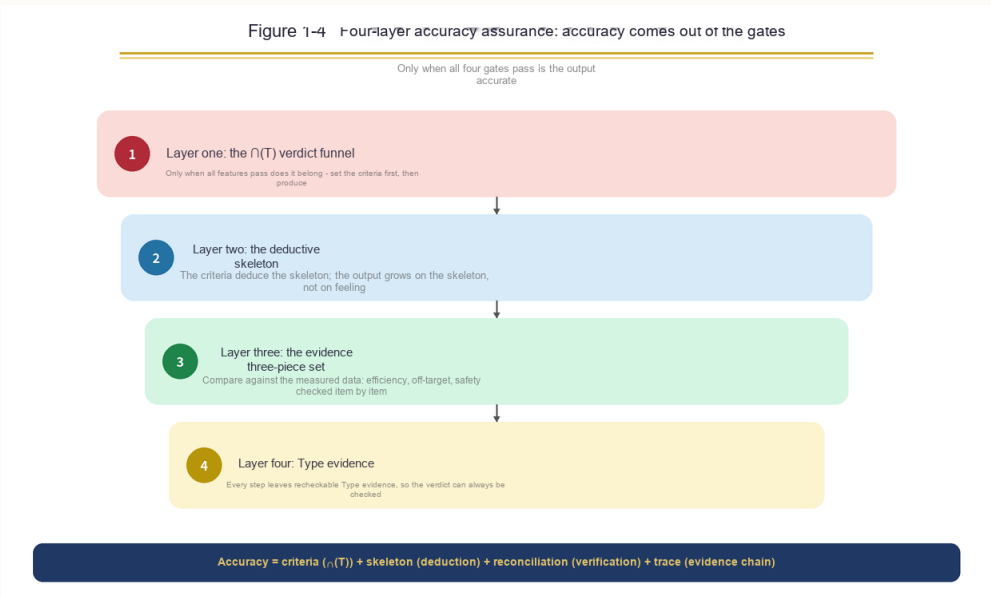


Figure 1-4 Four-layer accuracy assurance

1.5 Four Scenes of Inaccurate Output

It is clearest read backwards. Inaccurate gene R&D output almost always happens in four scenes: target drift, guide misalignment, missed off-target, verdict without basis. The four scenes share one root cause - the output was not written as a Type, and the layers were not written clearly.

Table 1-5 Four scenes of inaccurate output

Scene	Symptom	Root cause	Solution
Target drift	Target gene, variant site, and design goal each stated separately; the target is swapped midway and nobody notices	The target has no Type and no id(C)	The Target Type establishes identity
Guide misalignment	The sequence is right, the PAM is wrong, the editing window does not cover the variant site	Design conditions not written as ∩(T)	The Guide Type establishes the criteria
Missed off-target	Reporting efficiency but not off-target, leaving the safety assessment missing a whole block	Assessment fields incomplete	The Assessment Type has all five columns
Verdict without basis	The decision to enter lead optimisation is written in the meeting, and next quarter nothing can be checked	No Type evidence chain, no evolution ledger	Verdict Type + evolution ledger trace

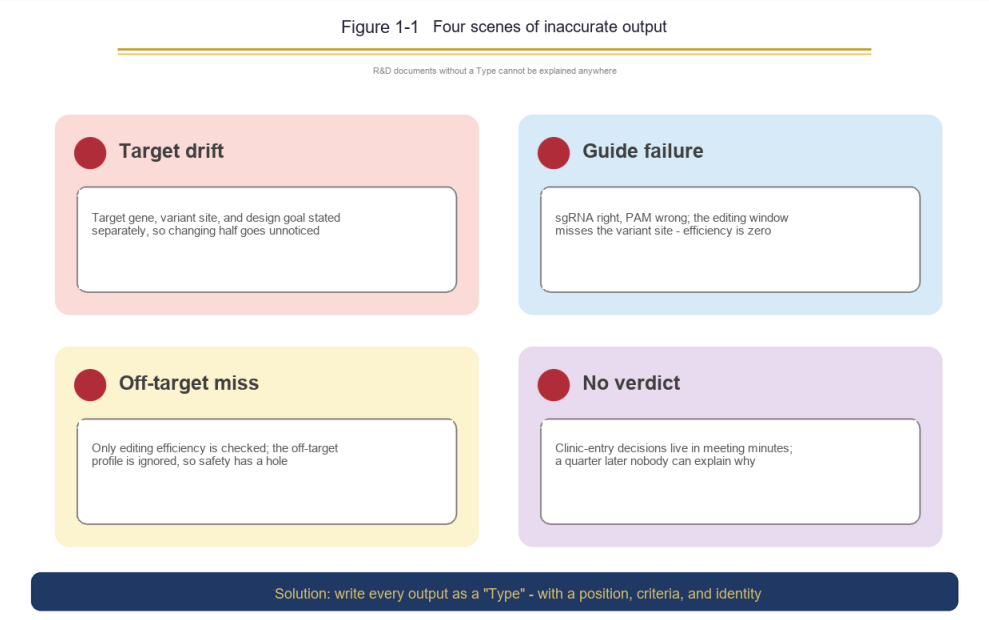


Figure 1-1 Four scenes of inaccurate output

Chapter focus

The AI gene R&D brain = a reasonable R&D twin: pack R&D into Type + Number, so every output has fields, criteria, and identity.

Where the two Types land: the three Space brackets write structure, the fourteen Time moves write operation; the Mind AI division of labour = LLM induces, Mind Model deduces.

Accuracy is gated out by four gates: criterion funnel, deductive skeleton, validation reconciliation, trace into a chain.

Chapter Two Six Output Types: "Target, Guide, Vector, Assess, Verdict, Evolution"



Explaining R&D relies on the output; explaining the output relies on the Type. The output chain of gene R&D has six Types: Target Type, Guide Type, Vector Type, Assessment Type, Verdict Type, Evolution Type - remembered as "target, guide, vector, assess, verdict, evolution".

2.1 Overview of the Six Types and the Output Chain

The six Types are not aliases for six documents but six sets of "field + criteria" specifications. The Target Type establishes the target, the Guide Type designs, the Vector Type delivers, the Assessment Type reconciles, the Verdict Type rules, and the Evolution Type closes - only when one chain is walked is one R&D run explained. Every Type is described with the same set of tools: [ ] writes field rules, { } writes composition, ( ) writes identity, ∩(T) writes criteria.

Table 2-1 Overview of the six output Types

Type	English	The question answered	Core criteria
Target Type	TargetType	Which gene to change? What disease to treat?	Variant clear + indication corresponding
Guide Type	GuideType	Which sequence and which editor are used to change it?	PAM qualified + window covering the variant
Vector Type	VectorType	How is it delivered into the target cell?	Delivery efficiency + safety meet the target
Assessment Type	AssessType	Is it changed accurately, and is it safe?	The three-piece set: efficiency, off-target, safety
Verdict Type	VerdictType	Lead optimisation or eliminate?	All four gates passed + nine cells filled
Evolution Type	EvolutionType	How to do better next round?	Iteration trace + criteria update

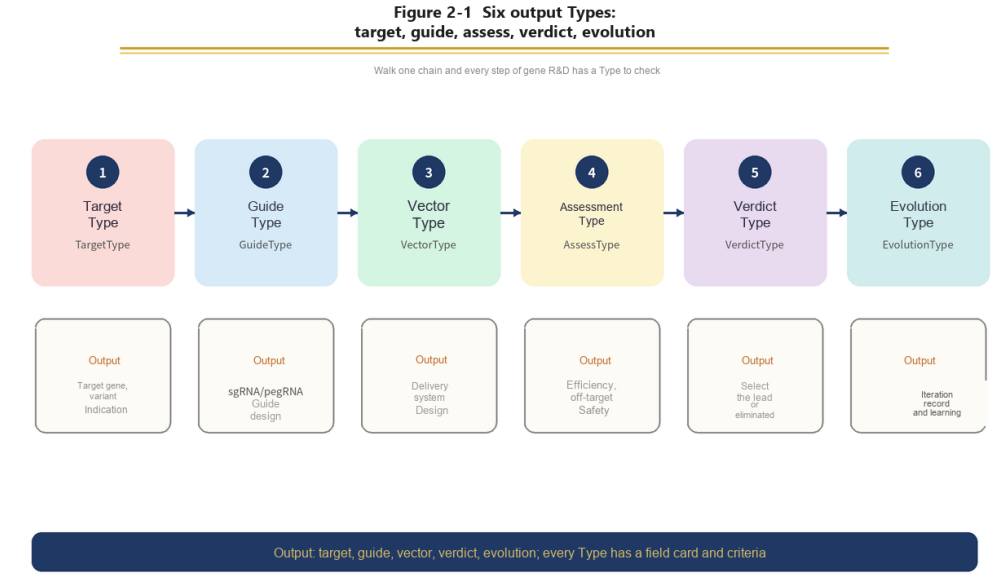


Figure 2-1 The output chain of the six output Types "target, guide, vector, assess, verdict, evolution"

2.2 Target Type and Guide Type

The Target Type is the anchor of the whole chain. Once the target gene, variant site, indication, and design goal are written, id(C) is established, and all later output hangs under this identity; need to change the target midway? Then open a new Target Type, not secretly edit a field. The Guide Type writes "how to change it" as a verifiable specification: complementary target sequence, PAM, editing window, editor choice - these four features form F(GuideType), and only when the intersection criterion ∩(T) fully passes is the guide qualified.

Table 2-2 Target Type field card

Field	Type	Criteria (∩(T))	Example (GX-2610)
Programme identity id(C)	Code	Unique, does not change with batch	GX-2610
Target gene	Gene symbol	Documented in OMIM/NCBI	HBB
Variant site	HGVS notation	c. notation + protein notation complete	HBB c.17A>T(p.Glu6Val)
Indication	Disease classification	Causal relationship with the variant supported by literature	Sickle-cell disease (SCD)
Design goal	Strategy	Correct, silence, or reactivate: one of three, with justification	Directly correct the point mutation
Variable content S(C)	Checklist	Explicitly list which fields can be swapped and which may not be touched	Batch cells and reagent lot numbers may be moved

Table 2-3 Guide Type field card

Field	Criteria	Inspection method	Example (PE-HBB-01)
Complementary target sequence	Fully complementary to the target site, 0 mismatches	Compare against the GRCh38 reference sequence	20 nt target sequence
PAM	The PAM required by the editor exists and is in the right position	Sequence scan	NGG(Cas9 nickase)
Editing window	Covers the variant site c.17	Window position comparison	Window +4 to +8 covers c.17
Editor choice	Matches the variant type (transition or transversion)	Strategy comparison table	Prime editing PE3max (can correct transversions)

Field	Criteria	Inspection method	Example (PE-HBB-01)
PBS/RTT parameters	The length falls within the empirical range	Literature + measured scan	PBS 13 nt/RTT 44 nt

2.3 Vector Type and Assessment Type

The Vector Type is the home ground of the BOM: { vector parts } writes the composition, and the three levels of remove write the impact. The Assessment Type is the ledger of accuracy: efficiency, off-target, safety, cell quality, phenotype recovery - five columns, each with a method, a threshold, and evidence; miss one and the verdict is missing a leg.

Table 2-4 Vector Type field card

Field	Criteria	Example (adopted in GX-2610)
Delivery platform	Matches the cell type, with clinical precedent	Electroporation (naked RNP + mRNA + pegRNA)
BOM composition	Only what recognises itself after removal enters the BOM	{ electroporation module, Cas9 nickase mRNA, pegRNA, buffer, targeting ligand (optional) }
remove grading	Every item graded at three levels	Targeting ligand = function-degraded part (still deliverable after removal, targeting degraded)
Delivery efficiency	Threshold: target cells >= 70%	82% (CD34+ cells, demonstration value)
Safety boundary	No insertional mutation risk, residues controllable	Non-viral route, no exogenous DNA integration

Table 2-5 Assessment Type field card

Field	Method	Threshold	Evidence
Editing efficiency	NGS sequencing quantification	Precise correction >= 40%	Sequencing raw files + analysis report
Off-target profile	GUIDE-seq + candidate site amplicon	Each off-target site < 1%	Off-target site list and quantification
Safety assessment	Survival rate, karyotype, large-fragment screening	Survival >= 80%, karyotype normal	Inspection report signature
Cell quality	Differentiation capacity, colony formation	CFU meets the target	Colony count record
Phenotype recovery	Erythroid differentiation culture, Hb phenotype	HbS falls significantly	HPLC electrophoresis profile

2.4 Verdict Type and Evolution Type

The Verdict Type writes "enter lead optimisation or eliminate" as a recheckable ruling: ruling, basis, conditions, and sign-off - all four columns present; the basis must point back to the four gates and nine cells, and the conditions must be acceptable. The Evolution Type is R&D's memory: what was learned, what was refactored, what was explored, and how the criteria were updated in each round are all recorded in the evolution ledger, and the next round grows on from the ledger.

Table 2-6 Verdict Type field card

Field	Content requirements	Common misuses
Ruling	Lead optimisation, eliminate, or supplement: one of three	Writing unverifiable phrases such as "agreed in principle"
Basis	Point back to the four-gate ∩(T) results and the nine filled cells	Citing only meeting conclusions, not data
Condition	Conditional pass acceptable, with a deadline and an owner	Conditions written as wishes (such as "complete as soon as possible")

Field	Content requirements	Common misuses
Sign-off	Three-party sign-off by R&D, QA, and biotech safety	Single-person sign-off with no second-person review

Table 2-7 Evolution Type field card

Field	Content requirements	Corresponds to moves
Learning	This round's data and literature enter the Type library	Continuous learning
Restructure	Which combinations to rearrange and re-measure	Refactoring and recombination
Explore	Alternative routes for parallel validation	R&D and exploration
Evolve	Update of criteria thresholds and Type definitions	Self-evolution
Version	Type library version number + change summary	Trace: who changed it, and why

2.5 The Three Brackets and the Three Levels of remove in R&D Use

Use the brackets in the wrong place and the layers collapse: [ ] is "what it should satisfy" - write the rules; { } is "what it is made of" - write the structure; ( ) is "where the identity is" - write the container. When discussing part impact, first pass the three levels of remove: identity lost after removal is an essential part, function missing but identity retained is a function-degraded part, no perceptible effect after removal is a non-essential part. Only when the three levels each sit in their place is the BOM complete.

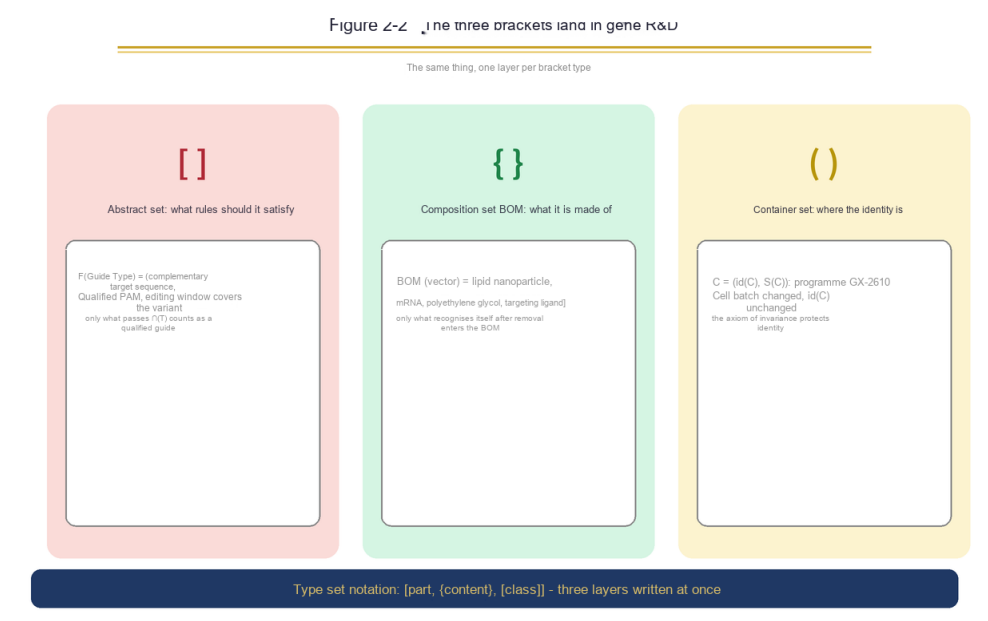


Figure 2-2 The three brackets land in gene R&D

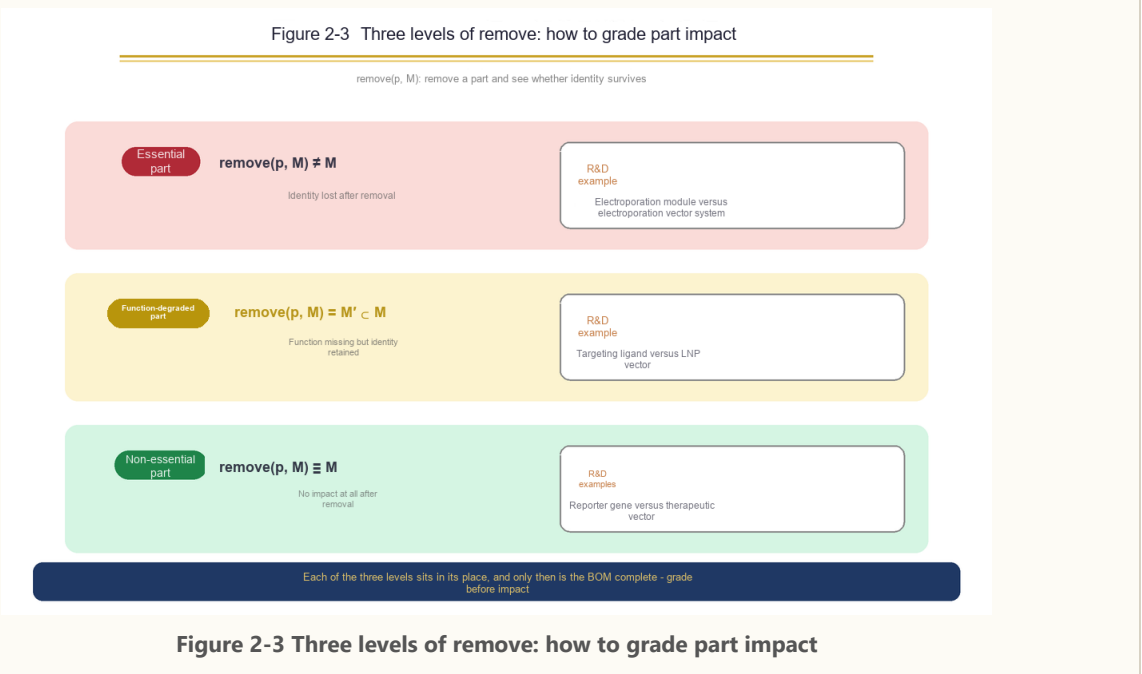


Figure 2-3 Three levels of remove: how to grade part impact

Table 2-8 Quick reference for BOM criteria and the three levels of remove

Criteria	How to write	R&D example
BOM criterion	What recognises itself after removal enters the BOM; what vanishes after removal enters the property	The electroporation module enters the BOM; "sterile" is a condition, not a part
Essential part	remove(p, M) ≢ M	Cas9 nickase mRNA versus the editing complex
Function-degraded part	remove(p, M) ≡ M′ ⊂ M	Targeting ligand versus LNP vector
Non-essential part	remove(p, M) ≡ M	Reporter gene versus therapeutic construct

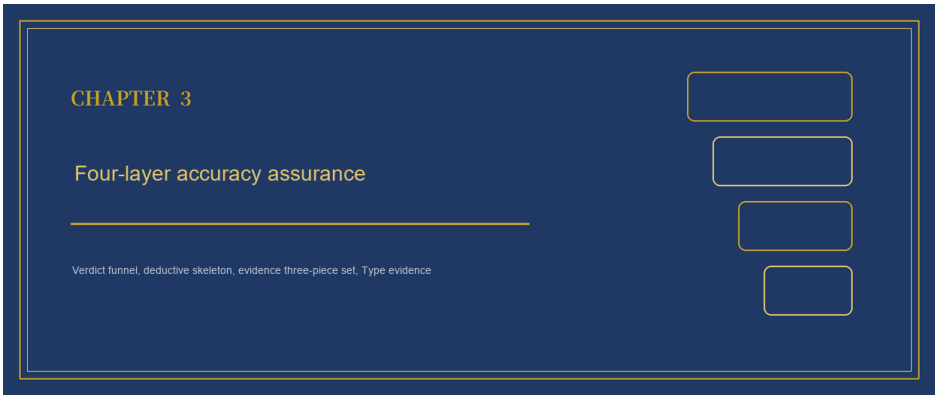
Chapter focus

The six Types "target, guide, vector, assess, verdict, evolution" each have a field card and criteria; only when one output chain is walked is one R&D run explained.

The Target Type establishes identity, the Guide Type establishes specification, the Vector Type establishes composition, the Assessment Type establishes the ledger, the Verdict Type establishes the ruling, the Evolution Type establishes memory.

The three brackets each in their place ([ ] rules, { } composition, ( ) identity), and the three levels of remove clarify part impact.

Chapter Three Four-Layer Accuracy Assurance



Accuracy is not an attitude but a structure. The four-layer accuracy assurance - criterion funnel, deductive skeleton, validation three-piece set, Type evidence chain - translates "the output must be accurate" into four operable, acceptable gates.

3.1 Layer One: the ∩(T) Criterion Funnel

Set the criteria first, then talk about output. Every Type first writes out its essential feature set F(T), and only when the intersection criterion ∩(T) fully passes does it count. In operation this becomes a funnel: a batch of candidates comes in and is filtered gate by gate, each gate with criteria, data, and handling - those that pass move on, those that fail are returned and leave a trace. The value of the funnel is repeatability: the same criteria, run by another person, give the same result.

Table 3-1 The four-gate ∩(T) funnel

Gate	Criteria	Evidence	Handling when it does not pass
Gate one features complete	Every field of F(T) complete and testable	Field card cross-checked field by field	Return and fill in the fields
Gate two criteria testable	Every criterion has a measurement method and threshold	Method literature + threshold basis	Return and re-establish the criteria
Gate three data reconciliation	Measured data pass the threshold item by item	Raw data + analysis report	Return for redesign or eliminate
Gate four trace on the chain	The first three gates' evidence is already in the chain	Type evidence chain nodes	Only after the chain is repaired may it be submitted for a verdict

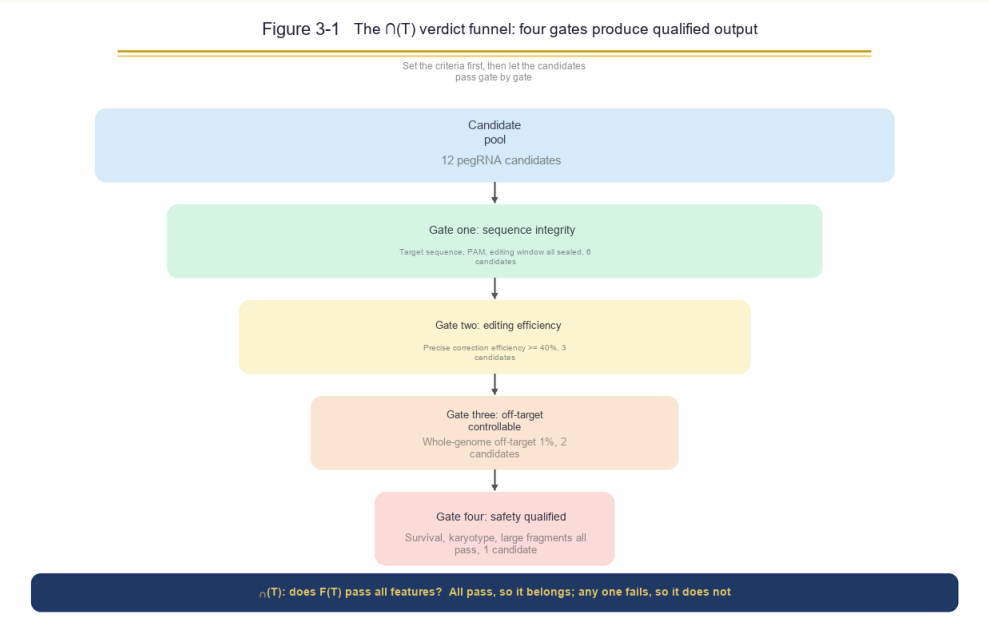


Figure 3-1 The ∩(T) criterion funnel: four gates filter out qualified output

3.2 Layer Two: the Deductive Skeleton

Output that passes the criteria must still grow on the skeleton. The deductive skeleton is the reasoning chain "cause, mechanism, evidence, method, validation": start from the cause and the variant, derive the pathogenic mechanism, set the evidence requirements, choose the editing strategy, and finally design the validation - one link clasping the next, with no link skipped. The time state is written as S(t+1) = g(S(t), move, event): every advance is driven by a move, and every state change leaves a trace.

Table 3-2 The deductive skeleton (cause, mechanism, evidence, method, validation)

Node	Deductive content	GX-2610 demonstration
Cause	What the cause and the variant are	HBB c.17A>T changes codon 6 of beta-globin from GAG to GTG, polymerising into sickle-shaped red blood cells
Mechanism	How the pathogenic mechanism proceeds	Deoxygenated HbS polymerises, red cells deform, vessels occlude and haemolysis follows
Evidence	What evidence counts	The three-piece set: precise correction rate, HbS phenotype reduction, off-target and safety
Method	Which strategy to choose to achieve it	Prime editing directly corrects the transversion (base editing only converts, so it does not apply)
Validation	How to verify that the strategy holds	In vitro erythroid differentiation culture + HPLC Hb phenotype + GUIDE-seq off-target

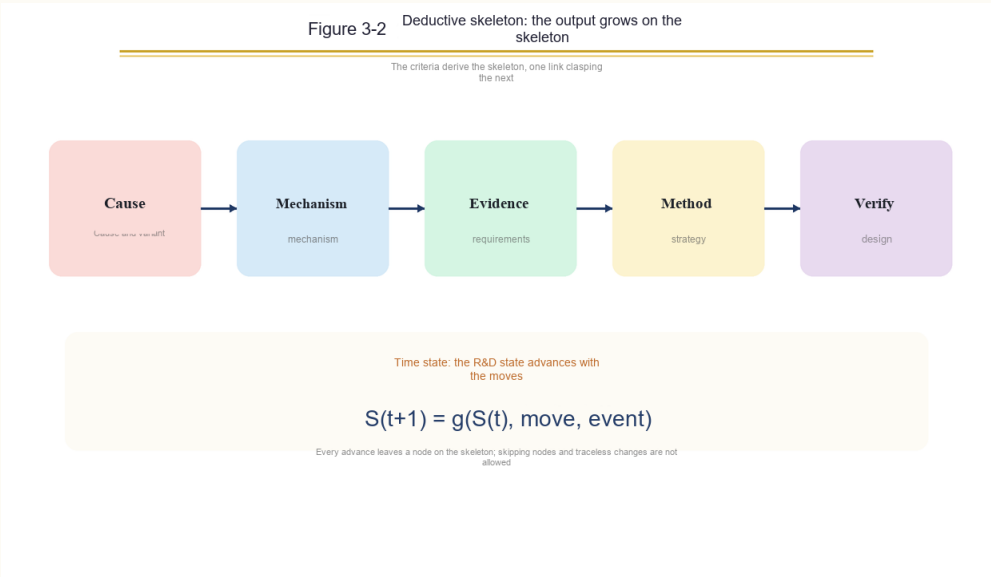


Figure 3-2 The deductive skeleton: the output grows on the skeleton

3.3 Layer Three: the Validation Three-Piece Set

The experiment is not the end of the process but the reconciliation point of the criteria. The validation three-piece set - in vitro efficiency, off-target analysis, safety assessment - each has a method, a threshold, evidence, and handling. Only when all three pass does the Type evidence chain connect; one missing and the chain breaks, and the verdict has no right to be signed.

Table 3-3 The validation three-piece set

Item	Method	Threshold	Reconciliation results and handling
In vitro efficiency	NGS sequencing quantification of precise correction rate and indel	Correction >= 40%, indel < 5%	Pass: enter the next gate; fail: return for redesign
Off-target analysis	GUIDE-seq whole genome + candidate site amplicon	Each off-target site < 1%	Pass: enter the next gate; fail: swap the guide or eliminate
Safety assessment	Survival rate, karyotype analysis, large-fragment deletion screening	Survival >= 80%, karyotype normal	Pass: submit for verdict; fail: supplement or eliminate

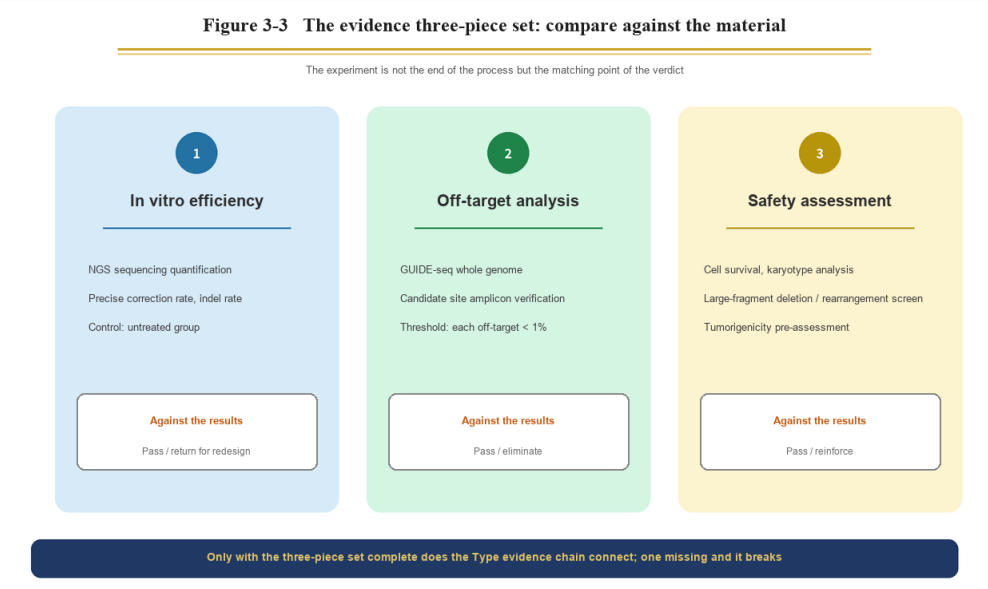


Figure 3-3 The validation three-piece set: the data come back to reconcile

3.4 Layer Four: the Type Evidence Chain

A verdict does not fear questions, it fears having no chain. The Type evidence chain strings one R&D run into six stations: requirement card, Type first draft, output documents, acceptance sheet, verdict document, evolution ledger - each station answers the same three questions: what were the criteria? Did it pass? Where is the evidence? If any station's chain breaks, all downstream output is returned; returning is not redoing the conclusion but reconnecting the evidence.

Table 3-4 The six stations of the Type evidence chain

Station	Output	Criteria	Evidence
1	Requirement card (S1)	Four framework steps completed, dimensions named	Situation description + dimension list
2	Type first draft (S2)	Six-Type fields opened, id(C) established	Field card draft
3	Output document (S4)	Fields of each Type filled and self-check passed	Six-Type output documents
4	Acceptance sheet (S5)	The four ∩(T) gates cross-checked one by one	Four-gate data reconciliation table
5	Verdict document (S6)	Nine cells filled + three-party sign-off	Sign-off record
6	Evolution ledger (S7)	All four columns - learning, refactoring, exploration, evolution - recorded	Version change summary

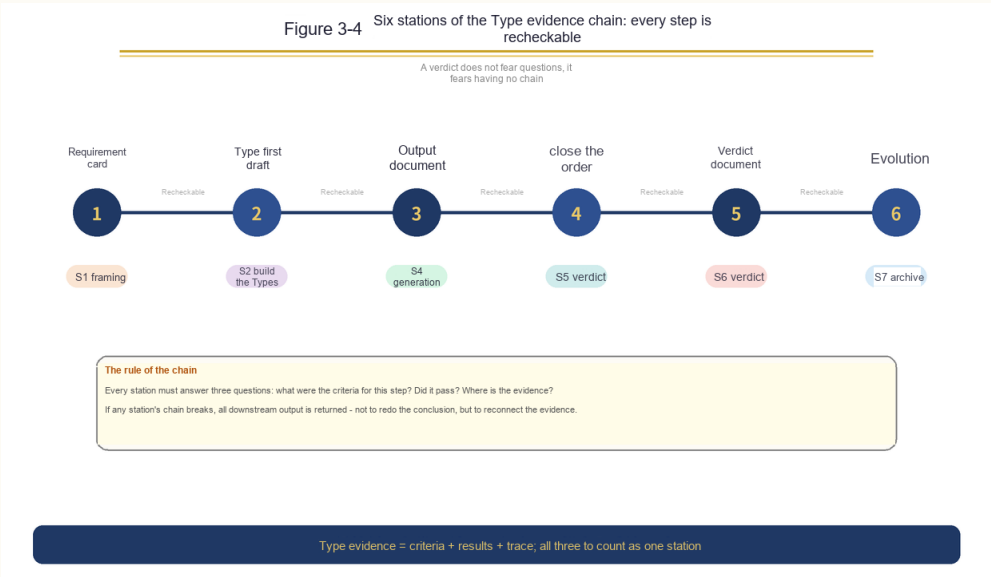


Figure 3-4 The six stations of the Type evidence chain: every step is recheckable

3.5 Cross-Acceptance: the Space × Time Nine-Cell Grid

The last gate is cross-acceptance: lay the three Space brackets (property, composition, container) and the three Time phases (design, validation, decision) into nine cells, one question per cell, one filled sheet per question. Only when all nine cells are filled and passed does the verdict document have the right to be signed - read horizontally it asks whether the structure is complete, read vertically whether the operation is solid.

Table 3-5 The nine cross-acceptance cells

	[ ] property layer	{ } composition layer	( ) container layer
Design phase	Are the features complete?	Is the part list complete?	Is the identity established?
Validation phase	Has it been measured item by item?	Has it been taken apart piece by piece?	Does id stay the same after the content is changed?
Decision phase	Have all criteria passed?	Has the impact been graded?	Is the verdict traceable?

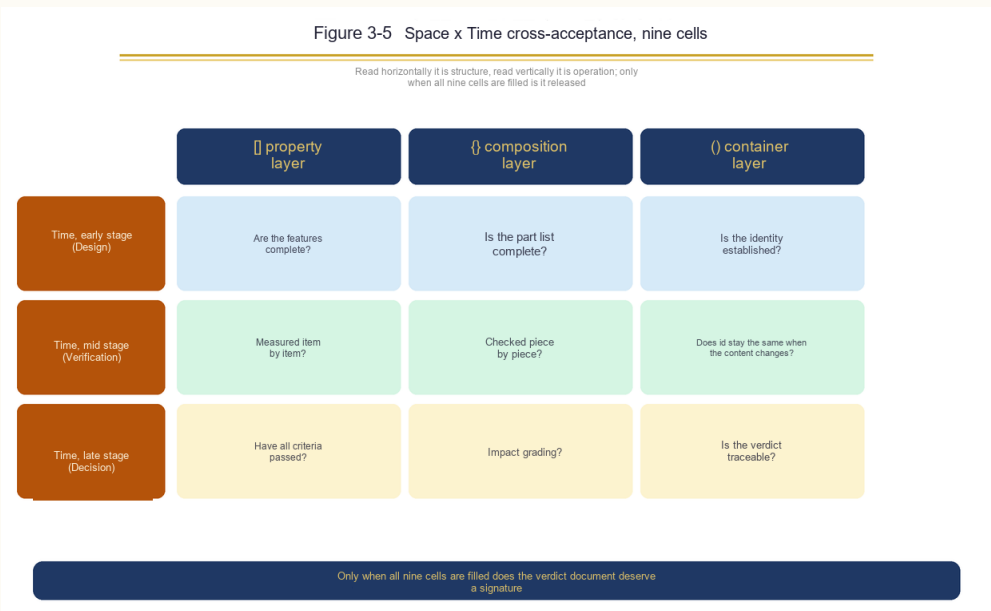


Figure 3-5 The Space × Time nine-cell cross-acceptance grid

Chapter focus

Four-layer accuracy: the ∩(T) criterion funnel, the deductive skeleton (cause, mechanism, evidence, method, validation), the validation three-piece set, the six stations of the Type evidence chain.

The experiment is the reconciliation point of the criteria: only when the three-piece set fully passes does the chain connect; one missing and the chain breaks.

Nine cross-acceptance cells: the three Space brackets x the three Time phases; only when all are filled may the verdict be signed.

Chapter Four Six Steps: from R&D Requirement to Output



Now that you know "what a Type looks like" and "how accuracy is guaranteed", this chapter strings the two into an operable process: the four framework steps identify the requirement, the six steps run the whole course, the three-layer judgement provides on-site control, and the move tray swaps moves with the situation.

4.1 Four Framework Steps: Identifying the R&D Requirement

An R&D requirement is not understood on receipt; it is forced out by "the control failed". The four framework steps - distinguish similarity and difference, trigger abstraction, name the dimension, judge membership - solidify this derivation into a repeatable mechanism: first distinguish "this is not that"; when the new difference does not line up with the old dimension, trigger abstraction; name the dimension and set the axis; finally judge the membership of each requirement. If it fits, add a scale; if it does not, open a new axis - the requirement card grows into a Type inside this loop.

Table 4-1 The four framework steps x organising the R&D requirement

Step	Operation	R&D example
One: distinguish similarity and difference	List the targets, diseases, and constraints in the requirement, and first distinguish "this is not that"	"Correcting the HBB point mutation" is not "reactivating HbF" - separate the two routes first
Two: trigger abstraction	When the new difference does not line up with the old dimension, admit a new dimension is needed	"Heritability" does not line up with any design dimension, so a new axis must be opened
Parameter, naming the dimension	Name the dimension and set the axis: target, delivery, efficiency, safety, regulation	Set the regulatory axis: germline editing forbidden, somatic cells through IND
Four: judging membership	Every requirement is mapped into a dimension; if it does not fit, open another axis	"Long-term follow-up" belongs to the regulatory axis and is not mixed into the safety axis

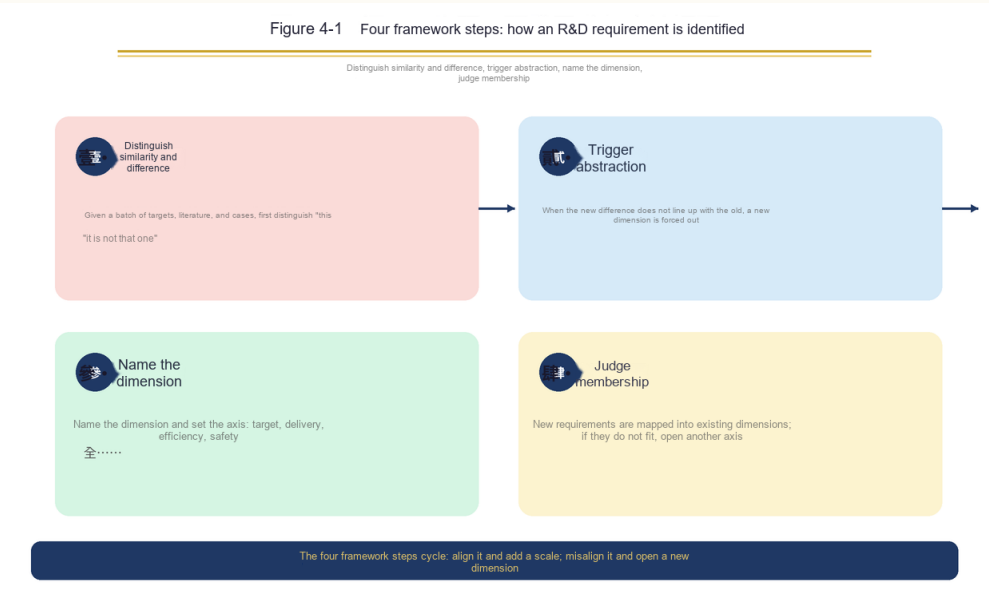


Figure 4-1 The four framework steps: how an R&D requirement is identified

4.2 Six-Step Operation Card

Once the requirement is identified, the six steps relay: identify the situation, build Types by axis, select the expansion pattern, generate the output, criteria acceptance, archive and evolve. Every step has an input and an output - the previous step's output is the next step's input; any step can be returned, but the return must be recorded in the evolution ledger, and traceless re-editing is not allowed.

Table 4-2 The six-step operation card

Step	Operation	Output	Commonly used moves
1 Identify the situation	The four framework steps organise the requirement	Requirement card	Problem solving, analysis
2 Build Types by Axis	The three brackets open the six-Type first draft	Type first draft	Design patterns
3 Select the Expansion Pattern	Select from the move tray + write the reason	Move selection sheet	Thinking methods
4 Generate the Output	Fill the fields Type by Type and generate the output documents	Six-Type output documents	Design, function generation
5 Criteria Acceptance	Four ∩(T) gates + nine cells filled	Acceptance sheet	Reasoning, analysis
6 Archive and Evolve	Enter the chain, keep the ledger, update the Type library	Evolution ledger	Continuous learning, self-evolution



Figure 4-2 The six-step operation card

4.3 Three-Layer Judgement: Is, Has, Right

Stop at any point in the process for the three-layer judgement: is - is the identity right? The container judgement asks "is identity independent of the contents"; has - is the structure complete? The composition judgement uses the three levels of remove to clarify impact; right - do the rules pass? The abstraction judgement runs ∩(T) and verifies on the spot. Only when all three pass is a conclusion drawn; if one layer fails, the conclusion is not a conclusion but a guess.

Table 4-3 Quick reference for the three-layer judgement

Layer	Question	Criteria	Example sentence from the site
Is	Is identity independent of the contents?	C = (id(C), S(C)), the axiom of invariance	Change to a batch of CD34+ cells - is GX-2610 still GX-2610?
Has	Are the parts complete and clearly graded?	BOM criterion + the three levels of remove	Is the targeting ligand marked as a degraded part?
Right	Has it passed ∩(T)?	Test F(T) item by item	Does this pegRNA pass all the features of the Guide Type?

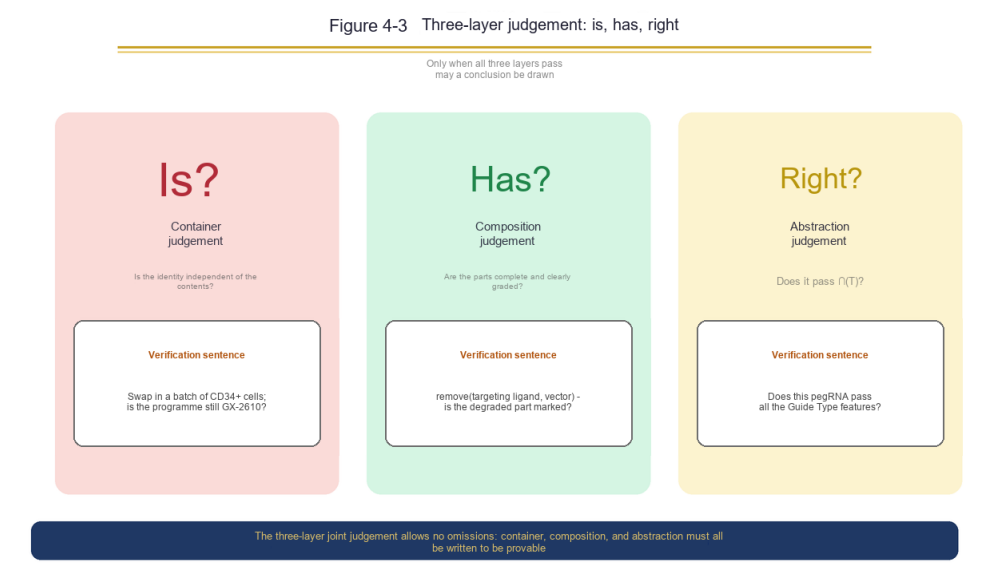


Figure 4-3 Three-layer judgement: is, has, right

4.4 Selecting, Assembling, Swapping Moves

The fourteen moves are a move tray, not a checklist. Select moves by situation: for pegRNA design pick "design + generation", for literature search pick "search + extraction", for risk assessment pick "analysis + reasoning". Several moves form a pipeline: the previous move's output is the next move's input, and the combination is written into the S1–S7 stages. If stuck, swap: when efficiency will not come up swap in "refactoring", when the assumption is unstable swap in "first-principles thinking" - both the reason for the swap and the result must be recorded in the evolution ledger.

Table 4-4 Situation table for selecting, assembling, and swapping moves

Action	Situation	Move combination	Trace
Selecting moves	Design pegRNA and editor selection	Design + logic generation	The move selection sheet states the reasons for selection
Selecting moves	Literature and patent research	Search + extraction + analysis	Source list entered in the Type library
Assembling moves	One complete R&D round	Search, analysis, design, validation, learning	S1–S7 embedded in the section
Swapping moves	The efficiency bottleneck cannot be broken	Problem solving, then refactoring and recombination	The evolution ledger records the reason for a move swap
Swapping moves	The strategy assumption wavers	Thinking methods (first principles)	The evolution ledger records the new hypothesis



Figure 4-4 Selecting, assembling, swapping moves

4.5 Misuse Warning Points

The process must run, and it must also avoid the reefs. Every station of the six steps has common misuse signals - handle the signal on the spot; it saves ten times the effort of remedying it later.

Table 4-5 Operating misuse warning points

Step	Warning signal	Immediate handling
Identify the situation	The requirement document cannot state three sentences of "this is not that"	Return to the four framework steps and first distinguish similarity and difference

Step	Warning signal	Immediate handling
Build Types by axis	The field card starts to become prose and the fields grow fewer	Stop writing and return to the field card, cross-checking field by field
Select the expansion pattern	The move selection sheet is blank or copied from last time	Re-select the move and write the reason; if no reason can be written, the wrong move was chosen
Generate the output	Outputs citing each other's conclusions	Pull it back to the skeleton: check link by link with cause, mechanism, evidence, method, validation
Criteria acceptance	The threshold is set after the data comes out	Void the criteria and re-establish them; lock the data before verifying
Archive and evolve	Two consecutive blank rounds in the evolution ledger	Only after the record is supplemented may the next round open

Chapter focus

The four framework steps (distinguish similarity and difference, trigger abstraction, name the dimension, judge membership) identify the requirement, and the six steps run the whole course.

The three-layer judgement is, has, right as a standing stop-check: only when identity, structure, and rule all pass is a conclusion drawn.

Moves follow the situation: select accurately, assemble smoothly, swap quickly; misuse warning points are handled on the spot, all leaving a trace.

Chapter Five The Full S1–S7 Case: GX-2610



This chapter connects all four previous chapters: it walks the S1–S7 seven-stage relay with the running case GX-2610 (HBB gene correction for sickle-cell anaemia), from the requirement card all the way to the evolution ledger - all values are demonstration data, demonstrating the output format and the operation of the criteria.

5.1 Seven-Stage Relay: the S1–S7 Pipeline

The seven-stage prompt relay is the standard usage of this book: S1 situation framing, S2 build Types by axis, S3 select the expansion pattern, S4 generate the output, S5 criteria acceptance, S6 cross verdict, S7 archive and evolve. Each stage's output is the next stage's input, and the handover between stages must carry the Type evidence; if any stage fails, return to the previous stage on the spot, and cross-stage jumping is not allowed.

Table 5-1 Overview of the seven-stage relay

Stage	Task	Input	Output
S1	Situation framing	R&D requirement, literature, constraints	Requirement card
S2	Build Types by axis	Requirement card	Six-Type first draft (field cards)
S3	Select the expansion pattern	Six-Type first draft	Move selection sheet
S4	Generate the output	Move selection sheet	Six-Type output documents
S5	Criteria acceptance	Output document + measured data	Four-gate acceptance sheet
S6	Cross verdict	Acceptance sheet + nine cells filled	Verdict document
S7	Archive and evolve	Verdict document + the whole chain	Evolution ledger + Type library version

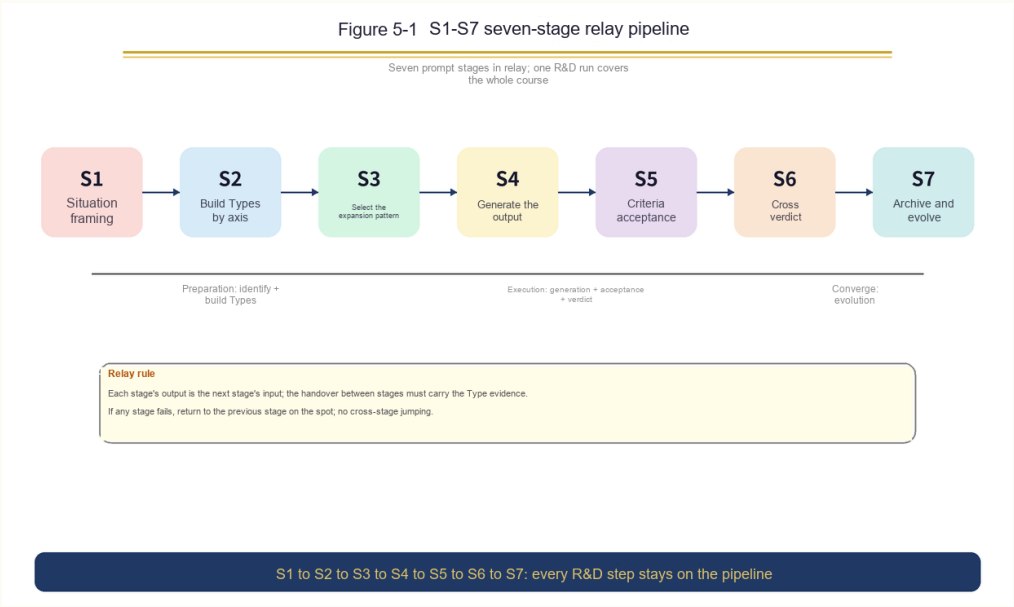


Figure 5-1 The seven-stage relay pipeline

5.2 Full Text of Prompts S1–S7

The following seven prompts can be pasted directly into the Mind AI platform; the text in brackets is the stage name, and the output format column is the basis for accepting that stage - if the output does not match the format, return and rerun; do not settle.

[S1 situation framing] Output: requirement card

You are the R&D planner of the AI gene R&D brain. Use the four framework steps to organise this R&D situation and output a requirement card:
(1) Distinguish similarity and difference: list the targets, diseases, and constraints, and write at least three sentences of "this is not that";
(2) Trigger abstraction: identify the new difference that cannot be explained by the existing dimensions;
(3) Name the dimension: name the new dimension (target / delivery / efficiency / safety / regulation ...);
(4) Judge membership: map each requirement into a dimension; if it does not fit, open a new axis.
Output format: requirement card = id(C) programme code + S(C) variable content list + dimension master table + open questions list.

[S2 build Types by axis] Output: six-Type first draft

Continue from the requirement card. Build the six-Type first draft with the three Space brackets:
[ ] Write the essential feature set F(T) and the intersection criterion ∩(T) for every Type;
{ } Write the BOM of each system, and check each item with "recognise itself after removal";
( ) Write the container C = (id(C), S(C)) and declare the axiom of invariance.
Output format: first draft of the six-Type field cards (Target Type, Guide Type, Vector Type, Assessment Type, Verdict Type, Evolution Type), one card per Type, with unfilled values marked "to be verified".

[S3 select the expansion pattern] Output: move selection sheet

Continue from the six-Type first draft. Select the moves this R&D run needs from the fourteen-move tray of the Time Type:
List the selected moves, the assembly order (the previous move's output = the next move's input), and the reason for selecting each move;
If the situation changes and a move must be swapped, write the swap conditions and the replacement move.
Output format: move selection sheet (move / purpose / input / output / swap conditions).

[S4 generate the output] Output: six-Type output documents

Continue from the move selection sheet. Generate the six-Type output documents in order: target card, guide card, vector card, assessment sheet, verdict document draft, evolution ledger opening.
Requirements: every item fills all fields of the field card; all values state their source (measured / literature / predicted); Undetermined values are marked "to be verified" and may not be left blank; each item ends with a self-check line: are all F(T) items testable?

[S5 criteria acceptance] Output: four-gate acceptance sheet

Continue from the output documents. Run the four-gate ∩(T) funnel acceptance:
Gate one features complete | gate two criteria testable | gate three data reconciled | gate four trace on the chain.
Each gate outputs "pass / return" plus the evidence metric; for a return, write the reason and the handling (supplement, redesign, eliminate).
Output format: four-gate acceptance sheet (gate / criteria / data / result / evidence location).

[S6 cross verdict] Output: verdict document

Continue from the acceptance sheet. Run the Space × Time nine-cell cross-acceptance ([ ] { } ( ) x the three phases of design, validation, and decision), filling each cell.
Only when all pass is the verdict document output: the ruling (one of three: enter lead optimisation, eliminate, supplement), the basis (citing the four gates and the nine cells),
Conditions (acceptable, with a deadline and an owner), sign-off columns (R&D, QA, biotech safety, three parties).
If any cell fails: return to the corresponding step; no verdict may be given directly.

[S7 archive and evolve] Output: evolution ledger

Continue from the verdict document. Output the opening of the evolution ledger:
Learning (this round's data and literature enter the Type library) | Refactoring (which combinations to rearrange) | Exploration (alternative routes) | Evolution (proposed threshold updates).
Write a version number and change summary for the Type library (who changed it, and why).
Output format: four evolution ledger columns + version note. The next round restarts from S1, and the requirement card must cite the evolution ledger.

5.3 Case Opening: Target Card and Guide Design

The GX-2610 situation: autologous CD34+ haematopoietic stem cells from a patient with sickle-cell anaemia, with the goal of directly correcting the pathogenic point mutation in the HBB gene. S1's requirement card sets the regulatory axis (germline editing forbidden, somatic cells follow the IND route), S2 opens the six-Type field cards, and S3 selects the combination "search + analysis + design + validation + learning". The following demonstrates the actual field entries of the S4 output document.

Table 5-2 Target Type card (GX-2610 demonstration)

Field	Value	Criteria check
id(C)	GX-2610	Unique, does not change with batch - check
Target gene	HBB (beta-globin gene)	OMIM 603903 is documented - check
Variant site	HBB c.17A>T(p.Glu6Val, rs334)	HGVS notation complete - check
Indication	Sickle-cell disease (SCD)	Causal literature on the variant sufficient - check
Design goal	Directly correct the point mutation (GTG to GAG)	Correction strategy: one of three, justified - check
S(C) variable content	Batch cells, reagent lot numbers, culture conditions	Movable fields listed explicitly - check

Strategy selection first passes one round of "selecting moves": base editing only converts (C to T, A to G) and cannot correct this transversion (A to T); HDR requires a double-strand break and has low

efficiency in stem cells; prime editing can install any point mutation - choose PE3max as the main line, with "disrupting the BCL11A enhancer to reactivate fetal haemoglobin" as a parallel control route. This reasoning, written on the move selection sheet, is the "method" node of the deductive skeleton.

Table 5-3 Guide Type card (pegRNA candidates, demonstration)

Code	PBS/RTT	nick site	Precise correction	indel	Keep or drop
PE-HBB-01	13 nt / 44 nt	+3	68%	2.1%	Enter assessment
PE-HBB-02	15 nt / 40 nt	+6	52%	1.6%	Enter assessment
PE-HBB-03	10 nt / 47 nt	−2	41%	2.8%	Enter assessment
PE-HBB-04	13 nt / 50 nt	+9	23%	3.4%	Efficiency gate returned

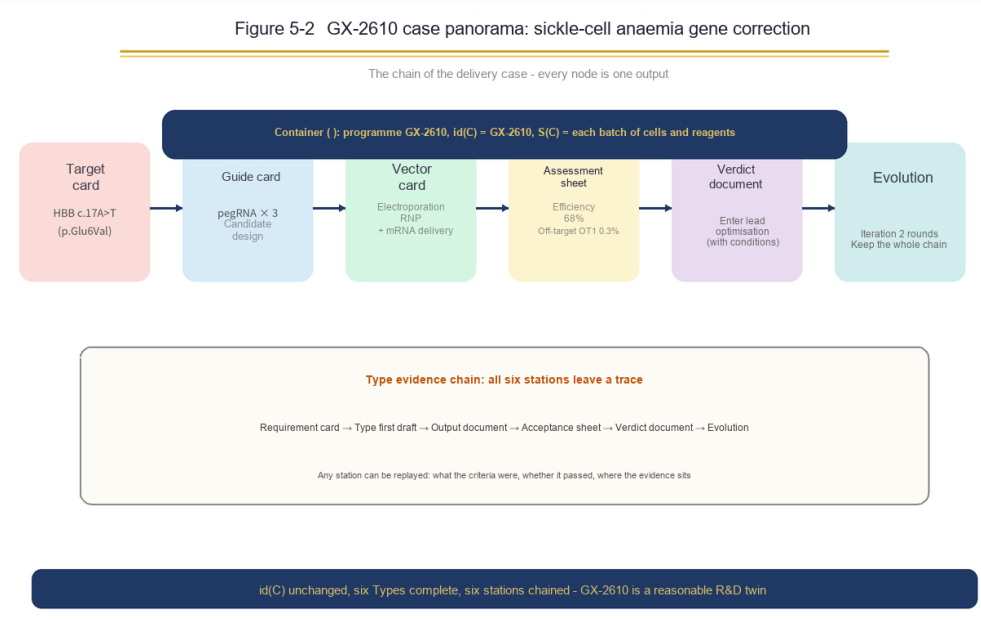


Figure 5-2 Panorama of the GX-2610 case

5.4 Full Output: Vector, Assessment, and Verdict

The vector card compares two delivery routes: electroporation (non-viral) and the AAV6 donor template. The basis for the decision is written on the field card - electroporation delivery efficiency 82%, no exogenous DNA integration risk, ample HSC clinical precedent, chosen as the main route; the targeting ligand is marked as a function-degraded part (still deliverable after removal, targeting degraded). The assessment sheet then enters the three-piece set data item by item.

Table 5-4 Vector Type card: route comparison (demonstration)

Field	Electroporation (adopted)	AAV6 donor template (control)
Delivery format	Naked RNP + Cas9 nickase mRNA + pegRNA	AAV6 viral vector carrying the donor template
Delivery efficiency	82% (CD34+, demonstration value)	65–75% (literature value)
Safety boundary	No exogenous DNA integration, residues controllable	Residual DNA must be tested; production cost is high
BOM	{ electroporation module, nickase mRNA, pegRNA, buffer, targeting ligand (optional) }	{ AAV6 particles, donor template, helper plasmid }
remove grading	Targeting ligand = function-degraded part	Donor template = essential part

Table 5-5 Assessment sheet: four-gate reconciliation (demonstration)

Gate	Criteria	Measured data	Result
Gate one sequence integrity	All pegRNA in vitro activities match	01/02/03 pass	Pass
Gate two editing efficiency	Precise correction >= 40%	01:68%|02:52%|03:41%	3 candidates pass
Gate three off-target controllable	Each off-target site < 1%	01:OT1 0.3%,OT2 0.1%|02:OT2 1.8%	01/03 pass
Gate four safety qualified	Survival >= 80%, karyotype normal	01: survival 86%, karyotype normal, no large fragments detected	01 comes through
Phenotype recovery (erythroid differentiation)	HbS falls significantly	HbS 41%→6%,HbA 0→76%,HbF 18%	Pass

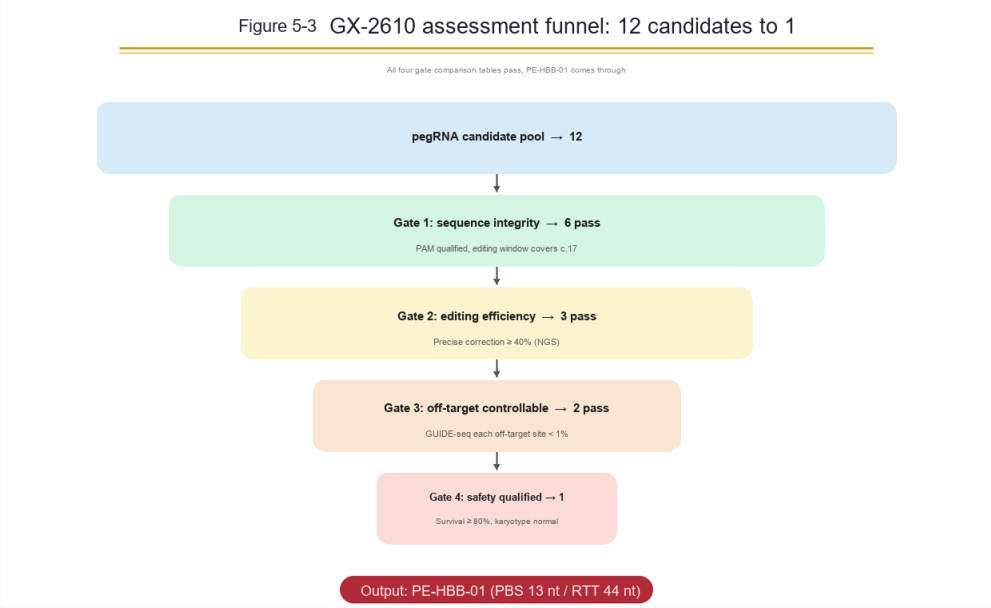


Figure 5-3 The assessment funnel: 12 candidates filtered to 1

Six cells and nine windows filled, four gates passed, the verdict document enters. Note the "conditions" column of the verdict document: a conditional pass is not a discount; it is writing the next round's acceptance criteria in advance.

Table 5-6 Verdict document (demonstration)

Field	Content
Ruling	Enter lead optimisation (conditional pass)
Basis	All four ∩(T) gates passed (PE-HBB-01); all nine cross-acceptance cells filled; the six stations of the Type evidence chain unbroken
Condition	(1) Complete in vivo assessment of in vitro off-target (deadline 60 days, owner: R&D team) (2) Schedule large-animal model validation (3) Bring long-term follow-up design into the clinical protocol
Sign-off	R&D lead | QA lead | biotech safety (three-party sign-off)

Disclaimer: the sequence parameters, efficiency, and off-target values in Tables 5-2 to 5-6 are demonstration data and do not constitute medical advice,
Nor is it a basis for real clinical trial design; real gene-editing therapy R&D must follow local regulations and ethics review.

5.5 Evolution Ledger and Usage Rules

After the verdict, the evolution ledger takes over. GX-2610 ran two iteration rounds in this demonstration: the first round converged parameters with a PBS/RTT length scan, and the second

optimised electroporation conditions to push efficiency from 52% to 68%; each round's four columns - learning, refactoring, exploration, evolution - are recorded in the ledger, and the criteria thresholds also update with the evidence - that is what "self-evolution" looks like concretely at the R&D site.

Table 5-7 Evolution ledger (two iteration rounds, demonstration)

Round	Learning	Refactor	Exploration	Evolution (criteria update)
Round 1	PBS 10-15 nt x RTT 40-50 nt scan: the 13/44 combination is optimal	pegRNA parameter space rearrangement	PE3max versus PE2	Efficiency threshold raised from 30% to 40%
Round 2	Electroporation voltage and pulse number affect survival and efficiency	Refactor the electroporation conditions and re-measure	Parallel observation of the HbF route	A new mandatory field "large-fragment deletion screening" added to the safety gate

Table 5-8 Rules for using the prompts

#	Rules	Reason
1	Establish the Type before acting: open the field card before generating content	Output without a Type cannot be accepted
2	Criteria first: the threshold is fixed before the data comes out	A threshold set afterwards is self-deception
3	Every output keeps the chain: the evidence location written into the acceptance sheet	A conclusion from a broken chain cannot be rechecked
4	Swap trace: the reason for the swap written into the evolution ledger	Traceless move swapping = decision without basis
5	The red line is the criterion: compliance clauses enter ∩(T)	Fail the red line and you fail the gate
6	Mark demonstration data: measured, literature, and predicted clearly separated	Mixed sources = numbers lose their meaning

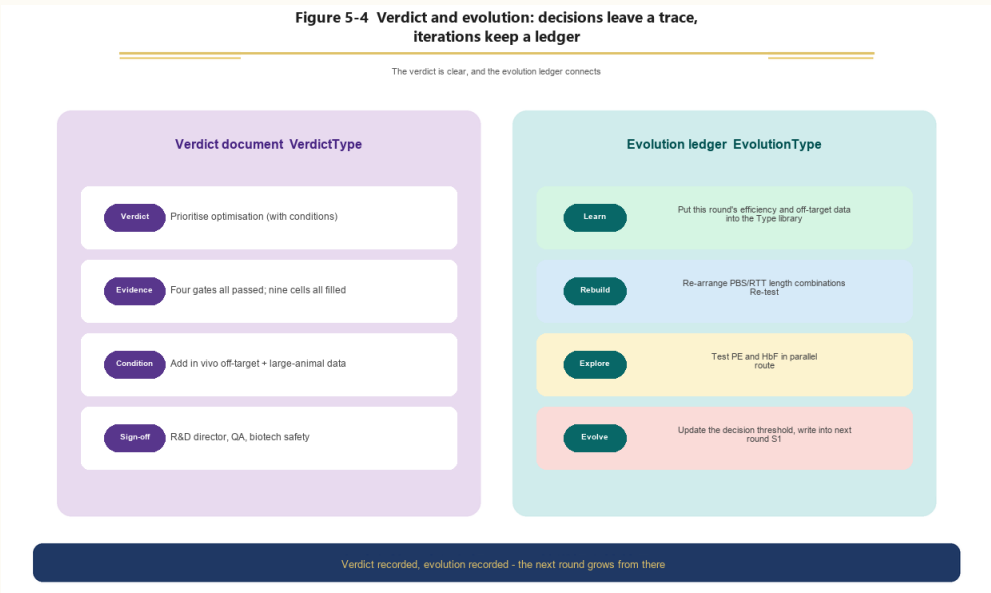


Figure 5-4 Verdict and evolution: decisions leave a trace, iterations have a ledger

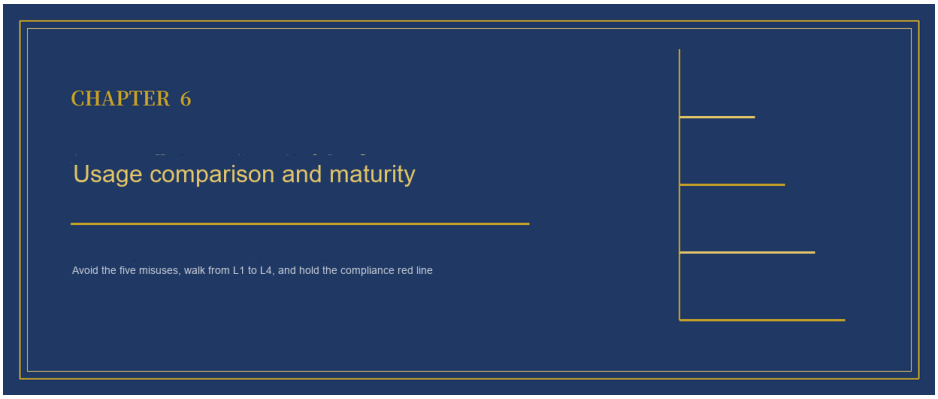
Chapter focus

Seven-stage relay S1–S7: situation framing, build Types by axis, select the expansion pattern, generate the output, criteria acceptance, cross verdict, archive and evolve.

Full GX-2610 demonstration: target card, guide card, vector card, assessment sheet, verdict document, evolution ledger - all six Types present, all four gates passed.

Six usage rules: establish the Type first, criteria first, keep the chain, leave a trace, the red line is a criterion, mark demonstration data.

Chapter Six Misuse Comparison and Maturity



The last chapter closes: avoid the five misuses first, carry the ten-question checklist, locate yourself on maturity L1–L4, schedule the ninety-day blueprint, and then draw the compliance red lines of gene editing clearly.

6.1 Five Misuses

The common root of misuse is only one: the layers were not written clearly. Substituting numbers for Types, using brackets at random, skipping grading, breaking stations on the chain, changing the verdict without a trace - the five pits each have their own scene, but all can be self-checked with the same sentence: is every output written as a Type? Is the right bracket used at every layer?

Table 6-1 Five misuses

Misuse	On-site symptom	Root cause	Correction
Substituting numbers for Types	The slides are full of percentages, and when asked "percentages of what" nobody can answer	Numbers are not hung on a Type	Back to the field card: label every number with its Type and field
Brackets used at random	[ ] { } ( ) written mixed, and the reader cannot grasp the layers	The semantics of the three brackets are not divided	Rules use [ ], composition uses { }, identity uses ( )
Skipping grading	Discussing part impact without the three levels of remove	Impact by feel	remove(p, M) graded item by item at three levels
Breaking a station on the chain	The acceptance sheet cannot find the corresponding criteria and evidence	The trace does not enter the chain	Any break is returned; connect the chain before discussing the conclusion
Traceless change of the verdict	The verdict was changed and the evolution ledger was not recorded, so nothing can be checked afterwards	Decision bypasses the trace	A criteria change must open a new version and record the reason



Figure 6-1 Five misuses

6.2 A Ten-Question Checklist

The ten-question checklist is a carry-along acceptance card: answer each question before submitting for a verdict, and whatever cannot be answered is a missing item. The checklist does not replace the four gates; it is a self-examination beyond the four gates.

Table 6-2 The ten-question checklist

#	Check question	Pass standard
1	Is the identity id(C) of the Target Type established?	Changing the target midway opens a new Type; it is not editing a field
2	Are the F(T) and ∩(T) of every Type written clearly?	Features testable item by item, no "roughly complies"
3	Are the three brackets in the right places?	[ ] rules, { } composition, ( ) identity each in its place
4	Does every BOM item pass "recognise itself after removal"?	Properties and parts are not confused
5	Have the three levels of remove been assigned?	Essential, degraded, and non-essential parts each have their place
6	Was the threshold set before the data came out?	The acceptance sheet timestamp is earlier than the data
7	Does the three-piece set (efficiency, off-target, safety) all have data?	No "to be completed" field enters the verdict
8	Are all nine cross-acceptance cells filled?	A blank cell means no verdict may be signed
9	Is the six-station Type evidence chain unbroken?	Three replay questions per station
10	Have the compliance red lines been checked one by one?	Red-line clauses entered ∩(T)

> **The last sentence before sign-off**
> Only when all ten questions pass is the verdict document pushed out. If any one cannot be answered - the cost of returning is a few days; the cost of signing wrongly is a programme.

6.3 Maturity L1–L4

Maturity is not a job title but the degree to which output can be rechecked. L1 can write a Type - the six-Type field cards can be opened; L2 can accept - the four gates run and the acceptance sheet matches; L3 can keep the chain - all six stations present and the verdict replayable; L4 can evolve - the evolution ledger drives iteration and the criteria thresholds update with the data. A reasonable goal for most teams is: from L1 to L2 within a quarter, and to L3 within half a year.

Table 6-3 Maturity L1–L4

Grade	Feature	Target metric
L1 Can write a Type	The output has fields and criteria	The whole team can open the six-Type field cards; three sentences can state "this is not that"
L2 Can accept	The four ∩(T) gates run	The acceptance sheet is one-to-one with the output documents; all thresholds precede the data
L3 Can keep the chain	All six stations of the Type evidence chain present	Any verdict can be replayed with the full reasoning within 24 hours
L4 Can evolve	The evolution ledger drives iteration	For three consecutive rounds the criteria thresholds update with the evidence; the Type library is versioned

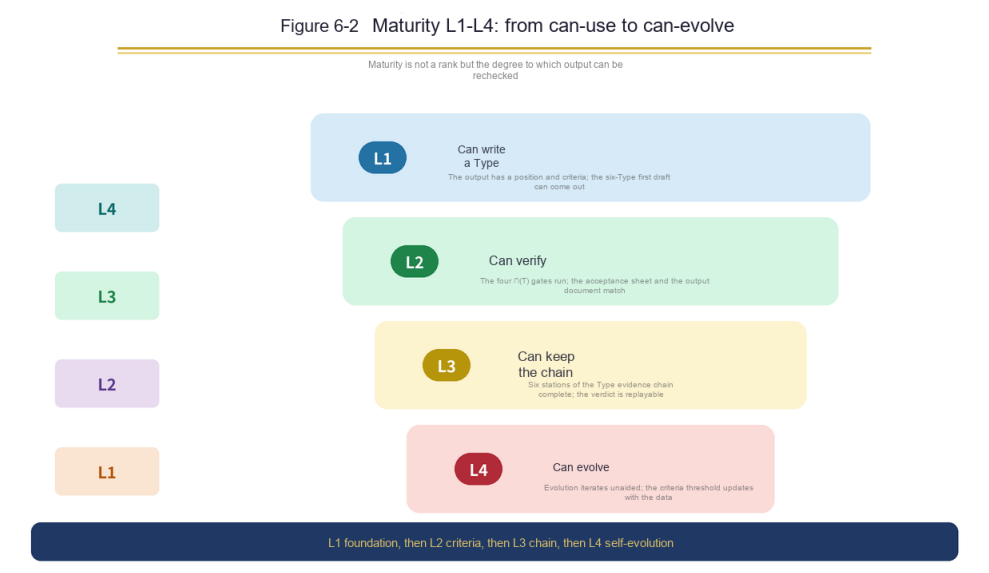


Figure 6-2 Maturity L1–L4

6.4 The Ninety-Day Adoption Blueprint

Adoption need not happen all at once. Three thirty-day blocks: in the first month establish the Types - six-Type field cards land, pick one programme for a trial run; in the second month establish the criteria - four-gate acceptance sheets and six-station Type evidence chain filing; in the third month establish the chain - the evolution ledger runs routinely and maturity is self-assessed. Each stage has acceptable milestones; if you miss one, step back and make up the lesson rather than pushing on.

Table 6-4 The ninety-day adoption blueprint

Stage	Milestone	Output	Acceptance
Days 1-30 establish the Types	Six-Type field cards land + one programme trial run	Field card template, S1–S7 trial run record	Everyone can open a six-Type first draft
Days 31-60 establish the criteria	Four-gate acceptance sheet + six-station filing	Acceptance sheet template, chain coding rules	One R&D round fully replayable end to end
Days 61-90 establish the chain	Evolution ledger routine + maturity self-assessment	Monthly evolution ledger report, L1 to L2 self-assessment form	The ten-question checklist brought into the sign-off process

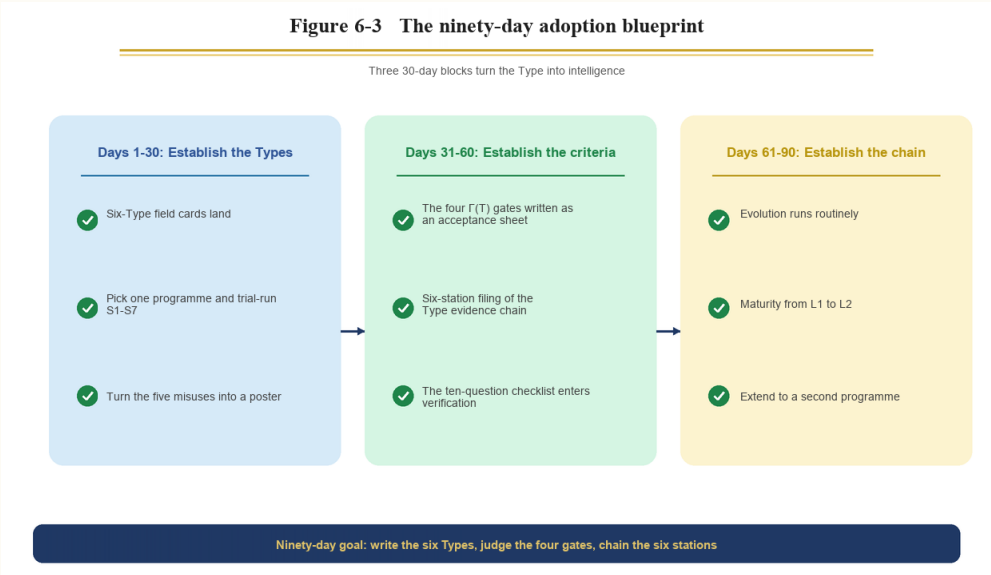


Figure 6-3 The ninety-day adoption blueprint

6.5 Compliance and Ethics Red Lines

The "accuracy" of gene editing also includes accuracy towards the red lines. The red line is not an appendix but part of the criteria - written into ∩(T); fail the red line and you fail the gate. Everyone must memorise four red lines: germline editing is forbidden to enter the clinical R&D process; somatic cell therapy follows the rules through IND and ethics review; long-term follow-up enters the Type evidence chain as a condition of the verdict; informed consent is an essential field of the Target Type.



Figure 6-4 Compliance and ethics red lines

Declaration: this section is a general compliance architecture demonstration and does not constitute legal advice; for actual R&D follow local regulations (such as human trials,
Gene therapy product management regulations) and the institutional ethics committee's review conclusions.

Chapter focus

The common root of the five misuses = the layers were not written clearly; one self-check sentence: is every output written as a Type?

Carry the ten-question checklist, locate yourself on maturity L1–L4, and schedule the ninety-day blueprint.

Compliance red lines are part of the criteria: germline editing forbidden, somatic cells compliant, long-term follow-up, informed consent.

Review: three takeaways

This book does only one thing: make the output of gene R&D accurate, testable, and recheckable. Before closing the book, you should hold three things.

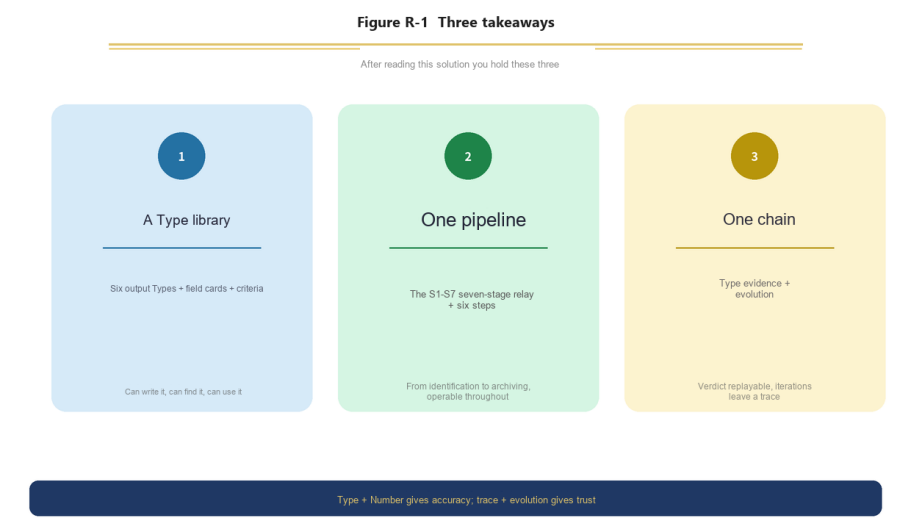


Figure R-1 Three takeaways

Table R-1 The book's toolbox

Tool	Where it is used	Section
Type + Number	Pack R&D into a reasonable twin	1.1
The three brackets [ ] { } ( )	Write the rules, write the composition, write the identity	1.2·2.5
The fourteen-move tray	Selecting, assembling, swapping moves	1.2·4.4
The six Types "target, guide, vector, assess, verdict, evolution"	The standard template of the output chain	Chapter 2
The four-gate ∩(T) funnel	The first gate of accuracy	3.1
Deductive skeleton (cause, mechanism, evidence, method, validation)	Let the output grow on the skeleton	3.2
The validation three-piece set	Reconciliation of efficiency, off-target, and safety	3.3
The six stations of the Type evidence chain	Every step recheckable	3.4
The nine cross-acceptance cells	The final check before signing the verdict	3.5
Four framework steps + six steps	The process from requirement to output	Chapter 4
Three-layer judgement (is, has, right)	On-site stop-check	4.3
S1–S7 prompts	Start directly at the seven-stage relay	5.2
The ten-question checklist, maturity, blueprint	Adoption and self-assessment	6.2–6.4

Appendix A Toolbox (quick reference)

Table A-1 Terminology quick reference

Terminology	English / notation	One-sentence definition
Type	Type	A fully described output (not a Form)
Type set	Type Set	Complete notation of a Unit Type: { part, (content), [class] }
Property, property value	Property·Value	Dimensions and scales
Abstract class	T = (F(T), ∩(T))	An open class determined by features and criteria
Intersection criterion	∩(T)	Only when all features pass does it count
Container	C = (id(C), S(C))	Set + invariant identity
Identity recognition	id(C)	A marker that does not change with the content
Axiom of invariance	Content replaced, id unchanged	Change the cell batch and the programme is still the same
Bill of Materials	BOM	The composition list when opened up
remove criterion	remove(p, M)	Remove the part and see whether identity survives
Three-layer judgement	Is, has, right	Joint check of container, composition, and abstraction
Move	function	Time Type = verb = function
Type evidence chain	Type Evidence Chain	The six-station chain of criteria + results + trace

Table A-2 The six Types x their main moves

Type	Main moves used	Supporting moves
Target Type	Analysis, problem solving	Search and extraction
Guide Type	Design (creation and innovation), design patterns	Logic generation
Vector Type	Design, function generation and application	R&D and exploration
Assessment Type	Analysis, reasoning	Statistical reasoning
Verdict Type	Reasoning (causal, simulation), thinking methods	Decision analysis
Evolution Type	Continuous learning, refactoring and recombination, self-evolution	First-principles thinking

Table A-3 Notation quick reference

Notation	Read as	Meaning
T = (F(T), ∩(T))	The definition of a Type	Feature set + intersection criterion
C = (id(C), S(C))	The definition of a container	Identity + internal space
{ part, (content), [class] }	Type set notation	Three layers of information written at once
remove(p, M) ≢ M	Essential part	Identity lost after removal
remove(p, M) ≡ M′ ⊂ M	Function-degraded part	Function missing, identity retained
remove(p, M) ≡ M	Non-essential part	No impact after removal
S(t+1) = g(S(t), move, event)	Time state advances	The R&D state is driven by moves
⇒	Implication	Pushed from criteria to conclusion

This solution document is a Mind AI x Ontypology application demonstration; the cases, values, and sequence parameters in the book are for demonstration.
Gene editing R&D involves regulatory and ethics review; follow local laws and institutional rules.



Type and Number united, genes
become judgeable

From target to verdict, every step leaves Type evidence

The LLM induces, the Mind Model deduces

AI Gene R&D Brain

MIND AI × ONTYPOLOGY